Split-Stream Hybrid Architecture for Language Models at Small Parameter Scales with High Memory Efficiency

Problem Statement

Modern artificial intelligence relies primarily on the transformer architecture, which requires significant computational resources. As models scale, systems with limited memory and resources struggle to deploy high-quality models. Improving model performance at small parameter scales while maintaining efficiency and quality is important for democratizing access. This project investigates a hybrid architecture that can improve performance while maintaining competitive memory usage

Research Questions

1. Can a hybrid architecture reduce peak GPU memory usage compared to a same parameter transformer?
2. Can a memory usage reduction be achieved without degradation in model performance, rather, an increase and betterment of the model?

Hypothesis

If a split-stream hybrid architecture model consisting of a state-space sequence model (SSM) and a Kolmogorov-Arnold Network (KAN), the model will improve performance faster relative to the same parameter-transformer while maintaining or reducing memory usage.

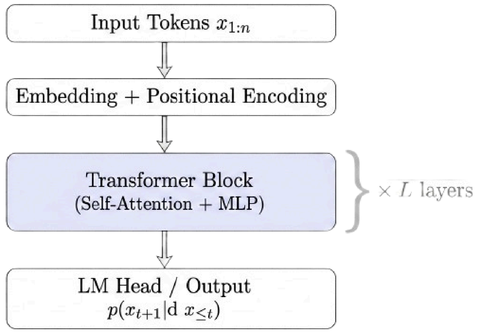
Background Information

- Over 212,000 AI companies currently exist, and most modern language models rely on the transformer architecture.
- State-space models provide efficient methods for modeling sequential data.
- Combining different sequencing techniques within a hybrid architecture may help leverage the strengths of both modeling approaches.
- Attention mechanisms capture relationships between tokens within a sequence, enabling language models to understand contextual dependencies.

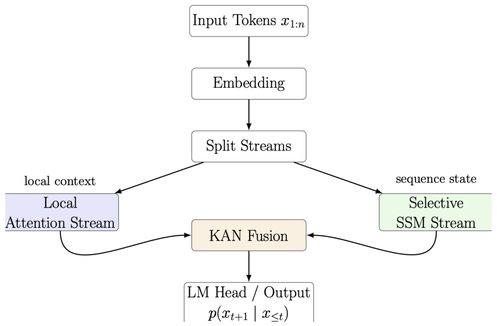
Key Idea + Importance

Instead of using a single processing pathway like a transformer, the proposed architecture processes sequences through two parallel streams and combines their representations through a fusion mechanism. This research suggests that hybrid architectures combining multiple sequence modeling approaches may improve model efficiency and enable more capable AI systems in environments with limited computational resources.

Model Architecture



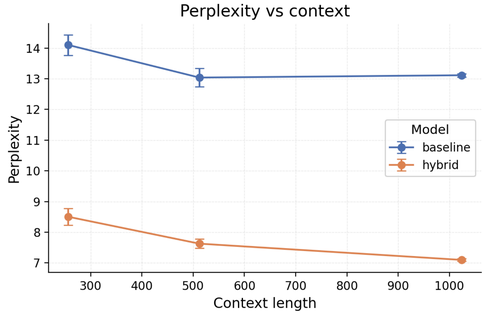
Baseline Transformer Processing Tokens Through a Single Pathway



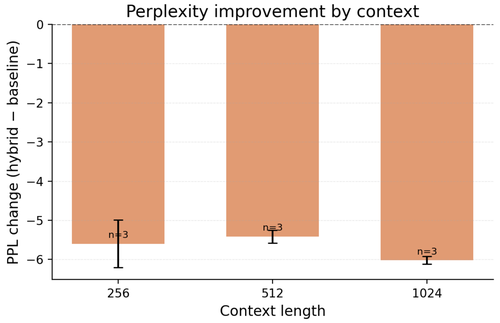
Hybrid architecture processes sequences through two streams and fuses them

Experimental Results

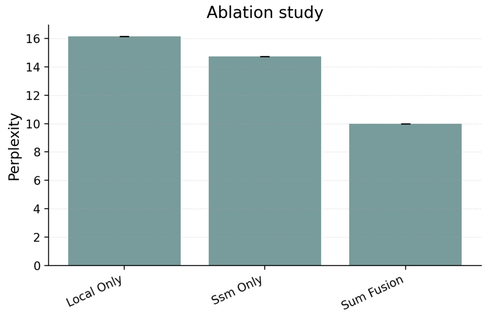
Improvement is consistent across three random seeds



Proves hybrid consistency outperforms baseline and gap continues to grow



Comparision Charts



This graph explains which component caused the steep drop in perplexity

Context length	Baseline PPL	Hybrid PPL	Improvement	Baseline tok/s	Hybrid tok/s
256	14.10 ± 0.33	8.50 ± 0.27	**39.70%**	12,698	63,910
512	13.04 ± 0.30	7.63 ± 0.15	**41.50%**	6,752	43,181
1024	13.11 ± 0.05	7.10 ± 0.04	**45.90%**	5,353	15,978

Context length	Quality gain	Memory overhead	Speed cost	PPL per 1K tok/s	Verdict
256	**−40%**	0.008	**−80%**	0.67	**Best quality gain**
512	**−42%**	0.001	**−84%**	1.13	**Lowest memory cost**
1024	**−46%**	0.0005	**−67%**	1.33	**Strongest overall**

Relevant Equations

$$f(\mathbf{x}) = f(x_1, \cdots, x_n) = \sum_{q=1}^{2n+1} \Phi_q\left(\sum_{p=1}^{n} \phi_{q,p}(x_p)\right),$$

Equation for KAN that was implemented throughout the project

$$\frac{ds}{dt} = -\lambda(t)s(t) + \lambda(t)b(t)$$

How a model's memory selectively forgets and absorbs information over time.
Integral solution powers our model

Procedure/Methodology

1. Implemented two models
   a. Transformer Baseline Model
   b. Split-Stream Hybrid Architecture Model
2. Ensured fair comparison by matching parameter counts, training conditions, and seeds,
3. Trained models across multiple context lengths
4. Evaluated performance by looking at validation perplexity (val_ppl), memory usage, and throughput

Result Interpretation

To ensure that the observed performance improvements were reliable and not caused by randomness during training, multiple controlled experiments were conducted. Both the transformer baseline and the hybrid architecture were trained under identical conditions, including matched parameter counts, datasets, and training schedules. Experiments were repeated using multiple random seeds to verify that the results were consistent across independent runs. The models were also evaluated across several context lengths to determine whether the performance trend persisted under different sequence conditions. These validation steps ensure that the observed improvements in perplexity are attributable to the architectural design rather than differences in training conditions or stochastic variation.

Conclusion

The results demonstrate that the split-stream hybrid architecture consistently achieves lower perplexity than the transformer baseline across all tested context lengths. This indicates that combining attention-based sequence modeling with state-space modeling can significantly improve language model prediction accuracy at small parameter scales. The ablation experiments further show that the fusion of the two modeling streams contributes most strongly to the observed performance gains. This suggests that the hybrid architecture successfully integrates complementary sequence modeling strategies to produce more accurate predictions. Overall, the findings support the hypothesis that hybrid architectures can provide meaningful improvements in language modeling performance while maintaining comparable computational characteristics.

Referenced Work

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